Midterm 2 review problems

These questions are NOT exhaustive.

1. An investigator is planning a study. They will have n iid samples of data X_i that are assumed to be Normally distributed with unknown mean μ and known variance $\sigma^2 = 2.5$. The null hypothesis is $H_0 : \mu = 3$ and alternate hypothesis is $H_1 : \mu \neq 3$. The investigator has budget to obtain $n = 10$ samples.

   (a) Provide and also sketch the distribution of the sampling distribution of $\bar{X}$ under the null hypothesis. Be sure to label the axes and add some x-axis values.

   (b) For a level 0.05 test, for what values of the test statistic $r(\mathbf{X}) = \bar{X}$ will the investigator reject the null hypothesis?

2. Let $X_1, \ldots, X_n | \theta \overset{\text{iid}}{\sim} \text{Unif}[\theta - 0.5, \theta + 0.5]$ with $\theta \in \mathbb{R}$ unknown.

   (a) Is the random variable $U = (\sum_{i=1}^n X_i) - n\theta$ pivotal? Why or why not?

   (b) Find a function $g(u, \mathbf{x})$ for which $g(U, \mathbf{X}) = \theta$.

   (c) Suppose $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \text{Unif}[-0.5, 0.5]$, let $Y = \sum Y_i$, and let F be the CDF for Y. Use the previous parts to find a formula for a γ-level confidence interval for θ in terms of F.

   (d) Suppose $n = 50$. What `R` code would you write to obtain/approximate the bounds of your interval?

3. Let $X_1, \ldots, X_n | \theta \overset{\text{iid}}{\sim} \text{Unif}[0, \theta]$, where $\theta > 0$ unknown. On a previous homework, you obtained an interval estimator for θ based on the pivot $U = \frac{Y}{\theta} \in [0, 1]$, where $Y = \max\{X_1, \ldots, X_n\}$. On that homework, you found the following:

$$F_U(u) = \begin{cases} 0 & \text{if } u < 0 \\ u^n & \text{if } 0 \leq u \leq 1 \\ 1 & \text{if } u > 1 \end{cases}$$

   I previously asked you to find a γ-coefficient confidence interval whose bounds are strictly greater than Y.

   (a) Now, I'd like you construct a different γ-coefficient interval estimator for θ of the following form: $[Y, B(\mathbf{X})]$. Find $B(\mathbf{X})$. *Try avoiding looking at your previous homework!*

   (b) For fixed γ, comment on if and how the length of your interval in (a) changes as $n \to \infty$. Is this behavior good or bad, and why?

4. Suppose we have a single $X | \theta \sim \text{Unif}[0, \theta]$. Consider a test of the following two simple hypotheses:

$$H_0 : \theta = 1 \qquad \text{vs.} \qquad H_1 : \theta = 2$$

   (i.e. $\Omega = \{1, 2\}$).

(a) Find a test that is level $\alpha_0 = 0$.

(b) Find the power of your test from (a). *Note: we want the* power, *not power function.*

(c) Suppose we have a test that rejects H_0 when $X \in [0, \alpha]$ for $0 < \alpha < 1$. What is the level and power of this test?

(d) Find another test with the same significance level and power as the previous.

5. Let $X_1, X_2|\theta \overset{\text{iid}}{\sim} \text{Unif}[\theta, \theta+1]$ for some $\theta \geq 0$, and $H_0 : \theta = 0$ versus $H_1 : \theta > 0$. We have two competing tests:

- δ_1: reject H_0 if $X_1 > 0.95$
- δ_2: reject H_0 if $X_1 + X_2 > c$ for some $c > 0$

Find the value of c such that $\alpha(\delta_1) = \alpha(\delta_2)$.

6. Suppose we have $X_1, \ldots, X_n|\theta \overset{\text{iid}}{\sim} \text{Bern}(\theta)$. Derive a (1-$\alpha_0$) $\times$ 100 confidence set for θ from the level-α_0 LRT of the hypotheses $H_0 : \theta = \theta_0$ versus $H_1 : \theta \neq \theta_0$.

7. A two stage clinical trial is planned to understand the efficacy of a certain protocol treatment. Let p be the proportion of patients in the treatment who respond to the treatment where it is known that $p \in [0.1, 1]$. Consider testing the following hypotheses:

$$H_0 : p = 0.10 \qquad \text{vs.} \qquad H_1 : p > 0.10$$

The two stage trial proceeds as follows:

- At the first stage, 10 patients are recruited and given the treatment. If 2 or more of these 10 patients are found to respond to the treatment protocol, then H_0 is rejected and the study is terminated.
- If the study was not terminated, we proceed to a second stage: an additional 10 patients are recruited and treated in the second stage. If a total of 4 or more patients among all 20 total patients respond to the treatment, then H_0 is rejected. No matter the results at this stage, the study is terminated.

For example: if 5 patients in the first stage are found to respond to the treatment, H_0 is rejected and the study is finished. However, if only 1 patient was responded in the first stage, we recruit another 10 patients. Suppose 1 additional person in stage two responded to the treatment. Since $2 < 4$, we fail to reject H_0. The study is terminated.

(a) Find the probability that H_0 is rejected at the first stage, assuming the null is true. State your answer exactly (using PMFs), and then write the `R` could you would use to obtain this probability.

(b) Find the overall probability that H_0 is rejected, assuming the null is true. State your answer exactly (using PMFs), and then write the `R` could you would use to obtain this probability.

8. Suppose $X_1, \dots, X_n | \mu, \sigma^2 \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$ where both μ and σ^2 are unknown. How should c be chosen such that the interval $(-\infty, \bar{X} + c]$ is a 95% confidence interval for μ?

9. Suppose we observe a single random variable $X|\theta \sim \text{Exp}(\theta)$. We would like to obtain a γ-coefficient confidence interval for the unknown shape parameter $\theta > 0$. Consider the following random variable: $Y = e^{-\theta X}$.

   (a) Find and state the exact distribution of Y.

   (b) Using (a), obtain a γ-coefficient confidence interval for θ that has equal-tailed probabilities. Be as specific as possible, and be sure to state your answer in the form of "[__, __]". *Note, you should be able to do most of this part even if you don't fully answer (a).*

   (c) Describe how to use part (b) to obtain a hypothesis test of level $1 - \gamma$ for the hypotheses $H_0 : \theta = \theta_0$ and $H_1 : \theta \neq \theta_0$.